

become a reality. These homes have electronic circuits and sensors embedded into their walls and floors and everywhere else and they allow you to operate your entire house from just a one point contact such as a smartphone/tablet app. These days many companies are working on assistive homes that can help elderly and disabled people lead a quality life without the help of institutional care. These homes are equipped with a set of wireless systems connected with a central context aware systems for monitoring daily schedules of the elderly people and using machine learning algorithms to learn and recognise behavioral patterns. This enables such systems to detect abnormal activities and react to emergencies relating to medical conditions.

Offices are also using the Embedded systems technologies. These days many organisations have Smart Offices that have floors with load sensing systems. These load sensing systems can detect any unusual load and notify the authorities.

The concept of context aware devices is also used to design Green or Zero Net Energy Buildings (ZNEB). These buildings use intelligent energy management systems using a set of sensors and actuators such as heat sensors, light intensity sensors and magnetic door sensors. These systems automatically lower down your energy usages by detecting the context information such as whether a device is being used or not, whether you are in the room or not etc. One such model is being designed by General Electric.

Personal assistants

Future embedded systems will see a wide range of applications in the area of personal assistance devices. Google already is working on GLASS, a glass shaped head mounted display that is just a hands-free ubiquitous computer that lets you interact itself using voice commands such as “Glass, Where am I?” and “Glass, Take a picture”. Such devices act as smart and intelligent personal assistants. EMIEW 2 is an office assistant robot developed by Hitachi. It acts as your workmate and follows you voice instructions such as “follow me”. Whenever it sees an object, it takes its picture and compares it with an online database of images. Swiss scientists at the Artificial Lab of the University of Zurich are working on the most advanced humanoid personal assistant or service robot Roboy. Roboy is funded by a Kickstarter campaign and is a hugely anticipated robot.